COLLEGE HELPDESK CHATBOT

Project Report — Rule-Based Chatbot Using Python if-else Conditions

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Course / Semester	Diploma in Information Technology / _____
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1. Introduction

A chatbot is a computer program that communicates with users through text or speech. This project is a simple College Helpdesk Chatbot. It uses a rule-based approach: the program checks the user's message for specific keywords and selects a response using Python if-elif-else conditions.

2. Objectives

- Create a basic chatbot using Python.
- Practise decision-making with if-elif-else conditions.
- Respond to common college-related questions.
- Handle unknown inputs politely.

3. Tools and Technologies

Language: Python 3
Interface: Command-line / terminal
Logic: Keyword matching, if-elif-else, functions, while loop
External libraries: None required

4. Working of the Chatbot

1. The program asks the user to enter a message.
2. The input is converted to lowercase and extra spaces are removed.
3. The get_response() function checks keywords in a sequence of if-elif-else conditions.
4. The matching response is printed on the screen.
5. The conversation continues until the user types 'bye', 'exit', or 'quit'.

5. Chatbot CODE

```
"""
College Helpdesk Chatbot
Topic: College information
Type: Rule-based chatbot using Python if-elif-else conditions

Run in terminal:
    python college_rule_based_chatbot.py
"""

def get_response(user_input):
    """Return a response by matching keywords in the user's message."""
    message = user_input.lower().strip()

    if not message:
        return "Please type a question so I can help."

    elif any(word in message for word in ["hello", "hi", "hey"]):
        return "Hello! Welcome to the College Helpdesk Chatbot. How can I help you?"

    elif "admission" in message or "apply" in message:
        return (
            "For admission information, check the college's official website or contact "
            "the admission office. Keep your marksheets, ID proof, and required documents
ready."
        )

    elif "fee" in message or "fees" in message or "payment" in message:
        return (
            "For the latest fee structure and payment deadlines, contact the accounts office
"
            "or check the official student portal."
        )

    elif "library" in message or "book" in message:
        return (
            "The library provides books and study resources. Please check the library notice
"
            "board or ask the librarian for current opening hours and borrowing rules."
        )

    elif "exam" in message or "timetable" in message or "schedule" in message:
        return (
            "Please check the official examination notice or student portal for the latest "
            "timetable, exam dates, and instructions."
        )
```

```
    elif "result" in message or "marks" in message or "grade" in message:
        return (
            "Results and marks are usually published through the official student portal. "
            "Use your authorized login details to check them."
        )

    elif "course" in message or "subject" in message or "syllabus" in message:
        return (
            "For course details and the latest syllabus, consult your department or the "
            "official academic notice/portal."
        )

    elif "scholarship" in message or "financial aid" in message:
        return (
            "Ask the student section about available scholarships, eligibility criteria, "
            "required documents, and application deadlines."
        )

    elif "contact" in message or "phone" in message or "email" in message:
        return (
            "Please use the contact details listed on the college's official website. "
            "I don't have verified current phone numbers or email addresses."
        )

    elif "thank" in message:
        return "You're welcome! Best wishes for your studies."

    elif "bye" in message or "exit" in message or "quit" in message:
        return "Goodbye! Have a great day."

    else:
        return (
            "Sorry, I don't have an answer for that question yet. Try asking about "
            "admissions, fees, library, exams, results, courses, scholarships, or contact
details."
        )


def main():
    print("=" * 55)
    print("       COLLEGE HELPDESK RULE-BASED CHATBOT")
    print("Type 'bye' or 'exit' to end the conversation.")
    print("=" * 55)
```

```
    while True:
        user_input = input("\nYou: ")
        response = get_response(user_input)
        print("Bot:", response)

        if user_input.lower().strip() in {"bye", "exit", "quit"}:
            break


if __name__ == "__main__":
    main()
```

5. Chatbot Conversations Output

```
PS D:\Collage\AI\ALA-3> python code.py
======================================================
       COLLEGE HELPDESK RULE-BASED CHATBOT
Type 'bye' or 'exit' to end the conversation.
======================================================

You: Hi
Bot: Hello! Welcome to the College Helpdesk Chatbot. How can I help you?

You: How do I get admission?
Bot: For admission information, check the college's official website or contact the admission office. Keep your marksheets,
ID proof, and required documents ready.

You: Tell me about the library
Bot: The library provides books and study resources. Please check the library notice board or ask the librarian for current
opening hours and borrowing rules.

You: When is the exam timetable?
Bot: Please check the official examination notice or student portal for the latest timetable, exam dates, and instructions.

You:
```

```
You: How do I check results?
Bot: Results and marks are usually published through the official student portal. Use your authorized login details to check
 them.

You: Can you tell me an unknown topic?
Bot: Sorry, I don't have an answer for that question yet. Try asking about admissions, fees, library, exams, results, course
s, scholarships, or contact details.

You: Bye
Bot: Goodbye! Have a great day.
PS D:\Collage\AI\ALA-3>
```

6. Algorithm

```
START
1. Display chatbot welcome message.
2. Read the user's message.
3. Convert the message to lowercase.
4. Check keywords using if-elif-else conditions.
5. Display the matching response.
6. If the user types bye, exit, or quit, end the program.
7. Otherwise, repeat from step 2.
STOP
```

7. Advantages and Limitations

Advantages: Simple to understand, fast, uses few resources, and is easy for beginners to modify.
Limitations: It only recognizes programmed keywords and may misunderstand spelling variations or complex questions. It does not connect to a live college database or verify current college information.

8. Conclusion

The College Helpdesk Chatbot demonstrates how a simple conversational program can be built with Python using conditions and keyword matching. It is suitable as a beginner rule-based chatbot project. Future versions could add a graphical interface, a database, or natural-language understanding.